Storage Lab: Stale NFS Mount Diagnosis & NetApp ONTAP SVM Export Policy Triage

Duration : 60 Min.

Scenario

Following an enterprise network migration, an application server reports 'ESTALE (Stale file handle)' whenever accessing `/mnt/netapp_shared`. The share is hosted on a NetApp ONTAP Storage Virtual Machine (SVM). Running `ls /mnt/netapp_shared` hangs indefinitely. The engineer must diagnose the stale mount, identify processes holding open file descriptors with `fuser`/`lsof`, perform a safe lazy unmount (`umount -l`), examine the NetApp SVM export policy rule set (`export_policy_dump.txt`) to discover that the client's new IP subnet (`192.168.20.0/24`) was missing from the export policy rule, formulate the remediation rule, remount successfully, and document the triage metrics in `stale_nfs_report.json`.

Task Objectives

Perform the following stale NFS mount recovery and NetApp triage tasks:

> **Pre-Staged Files in `student_workspace/`:**
> • **`export_policy_dump.txt`**: NetApp ONTAP SVM export policy rule dump.
> • **`stale_nfs_report.json`**: Starter stale mount diagnostic report with `TODO_*` placeholders.

1. Diagnose the stale mount state on `/mnt/netapp_shared` and identify processes holding open file handles using `fuser` or `lsof`.
2. Safely detach and unmount the hung filesystem using lazy unmount (`umount -l`) without causing kernel lockup or requiring a host reboot.
3. Analyze `export_policy_dump.txt` to identify the NetApp ONTAP export policy rule mismatch (`192.168.20.0/24`) causing access denial and specify the remediation command.
4. Execute the remount with corrected access parameters and verify active read/write transaction completion.
5. Record your diagnosed error code, safe unmount command, root cause subnet, and recovery status in `stale_nfs_report.json`.

Instructions to Perform the Task

1. Open your assessment **Terminal** or storage management CLI interface.

2. Execute the required storage management commands, volume operations, or configuration edits.
3. Update `stale_nfs_report.json` inside `student_workspace/` replacing all `TODO_*` placeholders with your verified configuration and diagnostic values.
4. Verify your storage configurations locally by executing:

```
python run.py
```

5. Once verified, return to the platform dashboard and click **"Run Tests"** / **"Full Eval"** to evaluate your score!

Validation

Once your storage tasks are executed, return to the platform dashboard and click the **"Run Tests"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Stale mount identification: accurately diagnose ESTALE error condition and identify processes holding locks	5 Marks
TC2	Safe unmount execution: successfully unmount the hung share using umount -l without leaving orphaned kernel states	5 Marks
TC3	NetApp export policy analysis: correctly identify the missing client IP in export_policy_dump.txt as root cause	5 Marks
TC4	Successful recovery: remount share successfully and verify read/write transaction completion	5 Marks

Total Score: 20 Marks

Important Notes

- This is an **Auto-Evaluated** storage assessment.

- Ensure all created volumes, files, and scripts are properly configured and saved before submitting.